

Rishi Soni

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Professional Summary

Computer Science undergraduate specializing in Artificial Intelligence (AI), Machine Learning (ML), Data Science, and Full Stack Web Development. Experienced in building scalable RESTful APIs, real-time applications, and production-grade ML pipelines using Python, Java, and JavaScript. Proficient with cloud platforms including Amazon Web Services (AWS) and Oracle Cloud Infrastructure (OCI). Holds 7 industry certifications across AWS, Oracle, and SAP. Strong background in deep learning, data engineering, and MLOps with hands-on experience in Docker, Kubernetes, and Apache Kafka.

Education

SRM Institute of Science and Technology *Expected Graduation: May 2027*
Bachelor of Technology (B.Tech) in Computer Science and Engineering | GPA: 8.85/10.0
Class XII – State Board *2022*
Percentage: 84.2%

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, TypeScript, HTML5, CSS3, SQL, PHP
Frameworks & Libraries: React.js, Node.js, Express.js, Django, FastAPI, Bootstrap, Streamlit, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, Google MediaPipe
Cloud & DevOps: Amazon Web Services (AWS), Oracle Cloud Infrastructure (OCI), Docker, Kubernetes, Firebase, CI/CD Pipelines, Linux
Data & ML Engineering: Apache Kafka, PySpark, Delta Lake, MLflow, XGBoost, Random Forest, Support Vector Machine (SVM), SHAP, A/B Testing, Feature Engineering
Databases & Tools: MongoDB, MySQL, PostgreSQL, Git, GitHub, Postman, VS Code, Jupyter Notebook, Socket.io, JWT Authentication
Other: RESTful API Design, Microservices Architecture, Responsive Web Design, Agile Development, Unit Testing, Debugging

Work Experience

Data Analyst Intern *Jul 2026 – Oct 2026*
Azlero Innovative Solutions (Remote)

- Developed SQL queries and data processing pipelines to extract, clean, and analyze business datasets for decision-making.
- Built dashboards and analytical reports while ensuring data accuracy, integrity, and confidentiality.
- Collaborated with cross-functional teams on data-driven projects, documentation, and reporting.
- Worked with business intelligence workflows, data validation, and trend analysis using industry-standard tools.

Software Engineering Intern

Summer 2025 – 6 Weeks

Defence Research and Development Organisation (DRDO)

- Contributed to secure software development projects following defense-grade coding standards.
- Collaborated with senior engineers on research workflows, technical documentation, and secure application development.
- Strengthened expertise in software architecture, secure coding, and analytical problem solving.

Projects

AI E-Commerce Recommendation System – Real-Time Collaborative Filtering Engine

Technologies: Python, PySpark, Apache Kafka, Delta Lake, MLflow, FastAPI, Docker, Kubernetes, Prometheus, Grafana

- Engineered a production-grade recommendation engine using collaborative filtering with SVD and NMF matrix factorization, achieving sub-100ms response latency, NDCG@10 of 0.78, and 94.2% user coverage.
- Implemented real-time event streaming with Apache Kafka and PySpark; integrated MLflow for model experiment tracking and deployed a FastAPI serving layer monitored via Prometheus and Grafana.
- Designed an A/B testing framework with statistical power analysis (power: 0.95), achieving a 23% click-through rate (CTR) lift (p-value: 0.001); containerized and orchestrated using Docker and Kubernetes.

AI Research Scientist Agent – Autonomous Multi-Agent Research Assistant

Technologies: Python, LangGraph, LangChain, OpenAI, FastAPI, Docker

- Developed an autonomous AI research assistant capable of literature review, paper summarization, hypothesis generation, and research planning.
- Implemented multi-agent workflows for search, reasoning, report generation, and citation management.
- Built REST APIs and modular agent pipelines supporting scalable AI-assisted scientific research.

Multimodal Parkinson's Disease Detection – Clinical AI Diagnostic System

Technologies: Python, PyTorch, ResNet-18, Wav2Vec 2.0, Scikit-learn, XGBoost, Librosa, SHAP, Streamlit

- Built a multi-modal machine learning system combining voice analysis and spiral drawing image classification to predict Parkinson's disease with 91.2% prediction confidence.
- Trained Random Forest, Support Vector Machine (SVM), and XGBoost classifiers on acoustic biomarkers (MFCC, pitch, spectral centroid, ZCR, chroma) extracted via Librosa; fine-tuned ResNet-18 CNN for motor-impairment image detection and integrated Wav2Vec 2.0 for raw audio embeddings.
- Implemented SHAP-based explainable AI (XAI) for feature importance visualization and deployed an end-to-end Streamlit dashboard supporting audio upload, clinical feature input, and spiral drawing analysis.

Static Application Security Testing (SAST) – AI Powered Code Security Scanner

Technologies: Python, FastAPI, AST Parsing, Docker, GitHub

- Built a static code analysis platform for detecting security vulnerabilities, insecure coding practices, and code quality issues.
- Implemented automated source code scanning and vulnerability reporting using rule-based analysis techniques.
- Designed REST APIs and containerized deployment for scalable security assessment workflows.

AI Mock Interview Platform – Intelligent Interview Preparation System

Technologies: Python, FastAPI, React, OpenAI, Speech Recognition

- Developed an AI-powered mock interview platform capable of generating technical interview questions and evaluating candidate responses.
- Integrated speech recognition and conversational AI for realistic interview simulations.
- Generated personalized performance reports and improvement suggestions using LLM-based evaluation.

ChatBox – Real-Time Full Stack Chat Application

Technologies: MongoDB, Express.js, React.js, Node.js (MERN Stack), Socket.io, JWT, REST API

- Developed a full-stack real-time chat application using the MERN stack with Socket.io for WebSocket-based bidirectional communication supporting concurrent users.
- Implemented JWT-based user authentication and authorization middleware, seen/unseen message tracking, and RESTful API endpoints for scalable back-end communication.

AI Fitness Tracker – Computer Vision Pose Estimation System

Technologies: Python, OpenCV, Google MediaPipe, Computer Vision, Pose Estimation

- Built a real-time AI fitness tracking application using Google MediaPipe and OpenCV for human body landmark detection and pose estimation from live video input.
- Computed joint angles using trigonometric analysis for posture evaluation and delivered real-time corrective feedback to improve exercise form accuracy.

FlyXpert – Full Stack Flight Booking and Management System

Technologies: HTML5, CSS3, JavaScript, PHP, MySQL, REST API, Admin Dashboard

- Developed a full-stack flight booking system with secure user authentication, real-time seat availability management, and RESTful API integration.
- Designed a responsive front-end UI and a role-based admin dashboard for flight and booking management using HTML5, CSS3, JavaScript, PHP, and MySQL.

Certifications

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| • AWS Certified Developer – Associate | <i>April 2026</i> |
| • AWS Certified Machine Learning Engineer – Associate | <i>April 2026</i> |
| • SAP Certified – SAP Generative AI Developer | <i>February 2026</i> |
| • Oracle Cloud Infrastructure 2025 Certified Data Science Professional | <i>September 2025</i> |
| • Oracle AI Vector Search Certified Professional | <i>September 2025</i> |
| • Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | <i>August 2025</i> |
| • Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | <i>August 2025</i> |